

Round three, Rural R&D for Profit summary of approved projects

On 4 May 2017, the Hon. Barnaby Joyce MP, the Minister for Agriculture, approved funding of more than \$35.8 million from 2016-17 to 2019-20 for seven projects under round three of the Rural R&D for Profit Program. This funding will be matched by more than \$59 million in cash and in-kind contributions by research and development corporations and their partners. The application period for the third round of funding opened on 6 October 2016, and closed 6 December 2016.

The successful projects will all contribute to the program objective – to realise productivity and profitability improvements for primary producers.

Project title	Applicant	Grant funding awarded (GST excl)	Applicant/ partner contributions (GST excl.)	Contributors	Summary
Wastes to profits: Technologies and business models for the management of wastes in the animal industries	Meat and Livestock Australia	\$6,000,000	\$4,651,826 cash \$3,721,903 in-kind	Australian Meat Processor Corporation; Dairy Australia Limited; Australian Pork Limited; Queensland University of Technology; University of Queensland; University of Southern Queensland; Murdoch University; Pacific Northwest National Laboratory; JBS Australia; Teys Australia; Australian Country Choice; Australian Lamb Company; Harvey Beef; Ridley Corporation Ltd; Aduro Biopolymers;	Australia's red meat, dairy, and pork industries produce significant quantities of wastes during on-farm production, intensive feeding and processing of animals. Waste management costs these industries more than \$200 million per year. This project will research and develop technologies and business models to reduce this cost as well as capture a market opportunity in excess of \$100 million per year by

					Zeolite Australia Pty Ltd; Barwon Water; Queensland Urban Utilities	converting wastes into valuable products such as fertilisers, feeds, chemicals and energy products for use in agriculture.
					Commonwealth Scientific and Industrial Research Organisation; Cotton Research and Development Corporation; Fisheries Research and Development Corporation; Ecological Australia; HVP Plantations; VicForests; Australian Bureau of Statistics; Australian Bureau of Meteorology; OneFortyOne Plantations	Natural capital is the soil, air, water and biodiversity – the natural resources used for food and fibre production. This project will apply Natural Capital Accounting to the forestry, cotton and fisheries industries allowing producers to:
Increasing farm gate profits, the role of natural capital accounts	Forest and Wood Products Australia (FWPA)	\$900,000	\$450,000 cash \$911,992 in-kind			
Improving plant pest management through cross industry deployment of smart sensor, diagnostics and forecasting	Horticulture Innovation Australia Limited	\$6,758,797	\$6,494,284 cash \$8,404,311 in-kind		Cotton Research and Development Corporation; Grains Research and Development Corporation; Wine Australia; Forest and Woods Products Australia; Rural Industries Research and Development Corporation (AgriFutures Australia); Plant Health Australia;	This project will develop a mobile, cross-industry plant pest surveillance network to monitor and report the presence of pests that threaten major agricultural sectors across Australia, including grains, cotton, sugar, horticulture, wine and forestry industries. The

Forewarned is forearmed: equipping farmers and agricultural value chains to proactively manage the impacts of extreme climate events	Meat and Livestock Australia (MLA)	\$6,198,942	\$3,660,370 cash \$4,803,123 in-kind	South Australian Research and Development Institute; VIC Department of Economic, Development, Jobs, Transport and Resources; Commonwealth Scientific and Industrial Research Organisation; WA Department of Agriculture and Food; Rothamsted Research; Nursery & Garden Industry Australia; B3Co\Plant & Food Research New Zealand; Sugar Research Australia Limited; AusVeg	surveillance network will cover:
				Grains Research and Development Corporation; Rural Industries Research and Development Corporation (AgriFutures Australia); Cotton Research and Development Corporation; Sugar Research Australia Limited; Wine Australia; Dairy Australia Limited; Australian Pork Limited; Australian Bureau of Meteorology;	Australian farmers and agribusiness operate in one of the most variable climates of any country in the world, with extreme events and climate variability the largest drivers of fluctuations in annual agricultural income and production. This project will deliver direct value to farmers through improving the seasonal forecast of

				University of Melbourne; University of Southern Queensland; QLD Department of Agriculture and Fisheries; Suncorp; Monash University; South Australian Research and Development Institute; VIC Department of Economic Development, Jobs, Transport and Resources	extreme climate events, including low and high rainfall, heat, cold and frost. It will equip farmers with the information and tools to be forewarned and change their management practices so they are proactively prepared. The project outputs will decrease the impacts of extreme climate events on farm and industry profit.
Boosting profit and reducing risk of mixed farms in low and medium rainfall areas with newly discovered legume pastures enabled by innovative management methods	Grains Research and Development Corporation (GRDC)	\$5,516,075	\$4,990,503 cash \$6,052,288 in-kind	Australian Wool Innovation Limited; Meat and Livestock Australia; Murdoch University; South Australian Research and Development Institute; WA Department of Agriculture and Food; Charles Sturt University; Commonwealth Scientific and Industrial Research Organisation	This project will develop recently discovered pasture legumes together with innovative management techniques to improve profitability for mixed farms (cropping and livestock) in the low and medium rainfall areas of WA, SA, Victoria and southern NSW. The new legume varieties will reduce nitrogen requirements, increase soil fertility, reduce weeds and diseases

High throughput technology for defining antimicrobial resistance status of pork and chicken meat enterprises leading to a competitive advantage in the global market place	Australian Pork Limited (APL)	\$1,274,000	\$648,286 cash \$999,157 in-kind	Rural Industries Research and Development Corporation (AgriFutures Australia); Murdoch University; University of Adelaide; Tecan Australia; Thermo Fisher Scientific; NSW Department of Primary Industries; Illumina	for following crops and be a source of quality feed for livestock. The project will also develop whole farm economic modelling to equip farmers with tools to adopt new pasture varieties and management practices, improving productivity and profits for grain, meat and wool producers. The project is expected to boost average farm profit by 10 per cent and halve economic risk. This project aims to provide Australian pig and chicken meat producers with a competitive advantage in the international marketplace by developing a world's best practice for objective description of the occurrence of antimicrobial resistance (AMR) at the herd/flock level. Using laboratory robots that can work extremely quickly, precisely and cost-effectively,
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the project will develop an industry-wide scheme where herds and flocks can be accurately described to define their level of AMR risk. This information will help producers demonstrate their AMR status, to not only improve market access but also provide a mechanism for monitoring progress of the on-farm control measures which will be based on antimicrobial stewardship principles.

Dung beetle ecosystem engineers – enduring benefits for livestock producers via science and a new community partnership model

Meat and Livestock Australia (MLA)

\$9,174,174

\$4,587,087 cash

\$8,932,275 in-kind

Commonwealth Scientific and Industrial Research Organisation; Invetec Pty Ltd; University of Western Australia; Landcare Research New Zealand; Mingenew Irwin Group; University of New England; Charles Sturt University; WA Department of Agriculture and Food; Dung Beetles for Landcare Farming; Dung Beetle Solutions

Dung beetles can improve the soil, reduce the spread of flies, pests and diseases, increase pasture health and reduce nutrient run off into waterways. This project will investigate how dung beetles can improve profitability and productivity for primary producers by:

Australia; Warren
Catchment Council;
Leschenault
Catchment Council

\$25,482,356

cash

\$33,825,049

Totals

\$35,821,988 in-kind

\$59,307,405

total

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/innovation/rural-research-development-for-profit/approved-projects-round3>